

İşaret Operasyonları (Signal Operations) Hafta4

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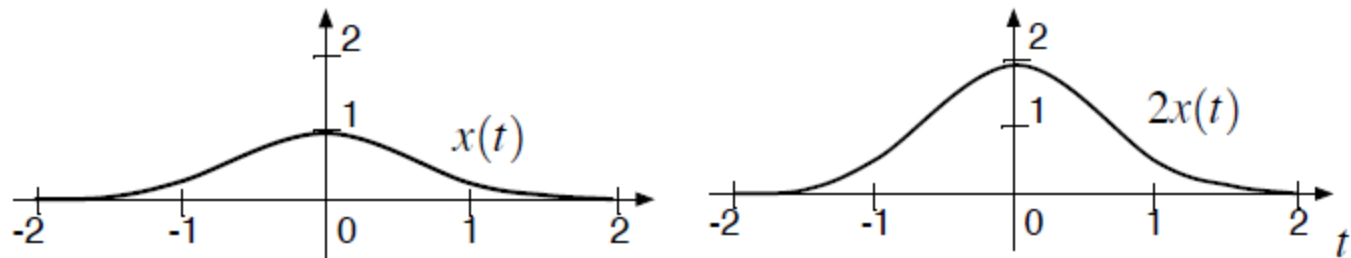
Textbook: Linear Systems and Signals, B.P. Lathi

This slides are modified from Linear Systems and Signals (EE313) course at the University of Texas at Austin, with Courtesy of Prof. Brian L. Evans

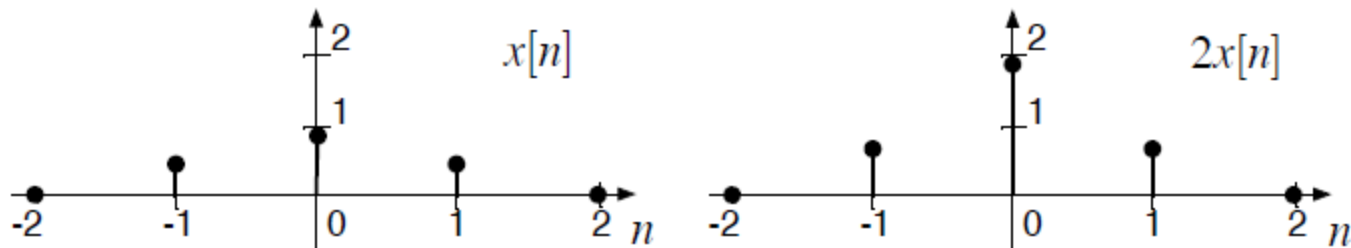
Amplitude Scaling

Amplitude Scaling

- The scaled signal $ax(t)$ is $x(t)$ multiplied by the constant a



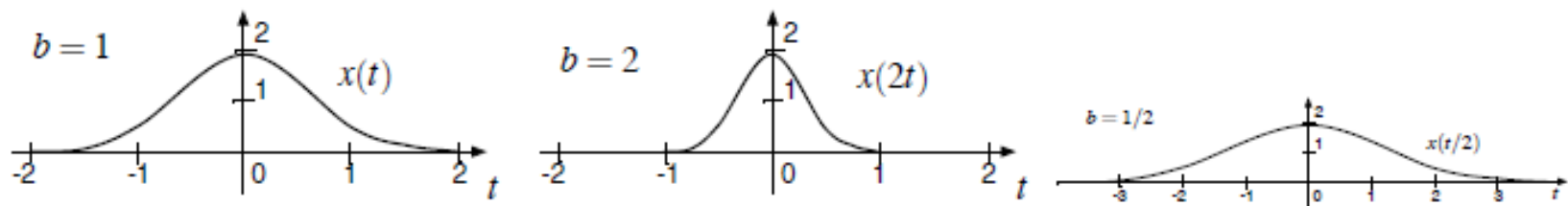
- The scaled signal $ax[n]$ is $x[n]$ multiplied by the constant a



Time Scaling

Time Scaling, Continuous Time

A signal $x(t)$ is scaled in time by multiplying the time variable by a positive constant b , to produce $x(bt)$. A positive factor of b either expands ($0 < b < 1$) or compresses ($b > 1$) the signal in time.

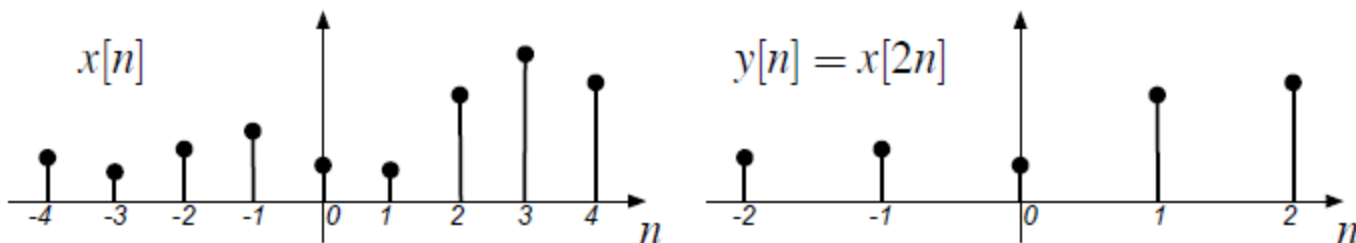


Time Scaling: Compression/downsampling

Time Scaling, Discrete Time

The discrete-time sequence $x[n]$ is *compressed* in time by multiplying the index n by an integer k , to produce the time-scaled sequence $x[nk]$.

- This extracts every k^{th} sample of $x[n]$.
- Intermediate samples are lost.
- The sequence is shorter.

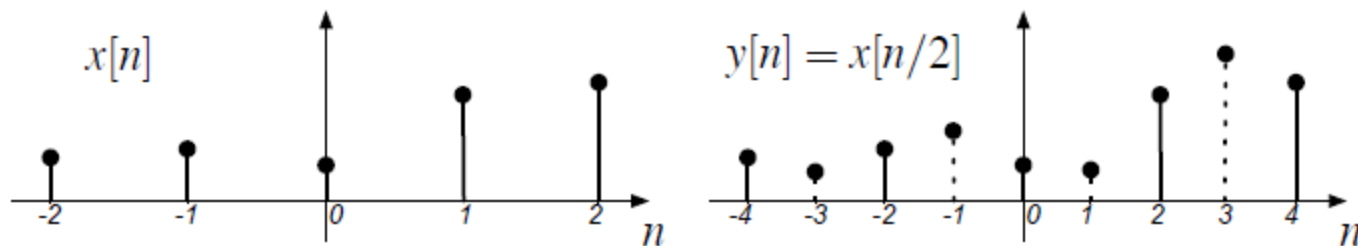


Called *downsampling*, or *decimation*.

Time Scaling: Expansion/Upsampling

The discrete-time sequence $x[n]$ is *expanded* in time by dividing the index n by an integer m , to produce the time-scaled sequence $x[n/m]$.

- This specifies every m^{th} sample.
- The intermediate samples must be synthesized (set to zero, or interpolated).
- The sequence is longer.

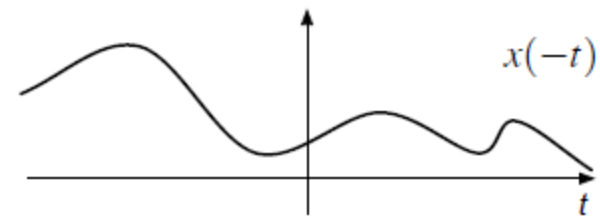
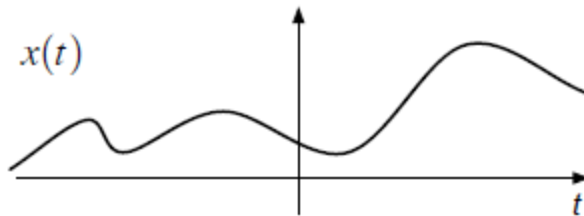


Called *upsampling*, or *interpolation*.

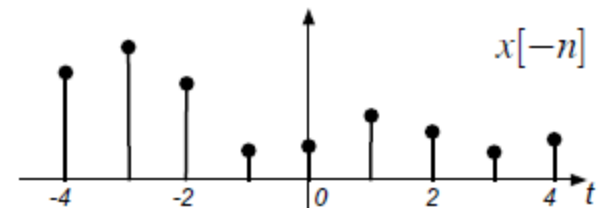
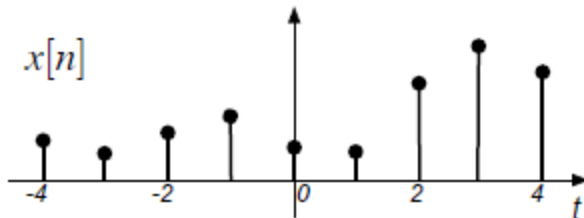
Time Reversal

Time Reversal

- Continuous time: replace t with $-t$, time reversed signal is $x(-t)$



- Discrete time: replace n with $-n$, time reversed signal is $x[-n]$.



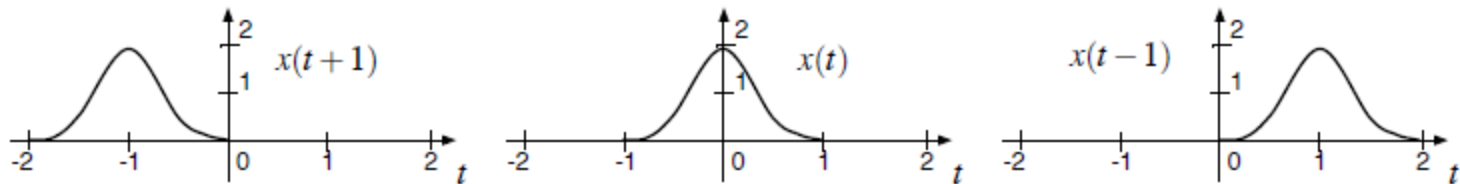
- Same as time scaling, but with $b = -1$.

Time Shift

Time Shift

For a continuous-time signal $x(t)$, and a time $t_1 > 0$,

- Replacing t with $t - t_1$ gives a *delayed* signal $x(t - t_1)$
- Replacing t with $t + t_1$ gives an *advanced* signal $x(t + t_1)$

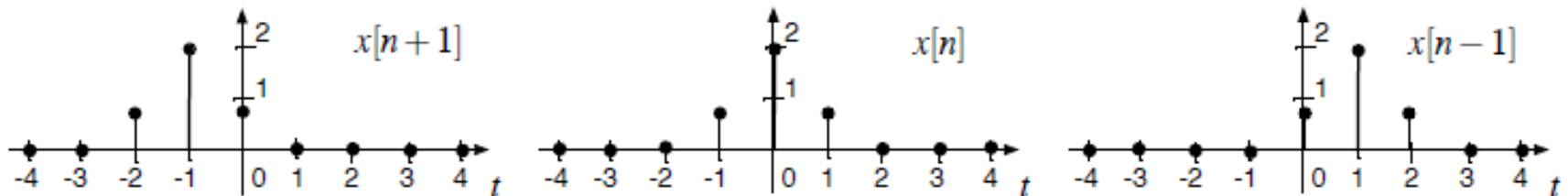


- May seem counterintuitive. Think about where $t - t_1$ is zero.

Time Shift

For a discrete time signal $x[n]$, and an integer $n_1 > 0$

- $x[n - n_1]$ is a delayed signal.
- $x[n + n_1]$ is an advanced signal.
- The delay or advance is an integer number of sample times.



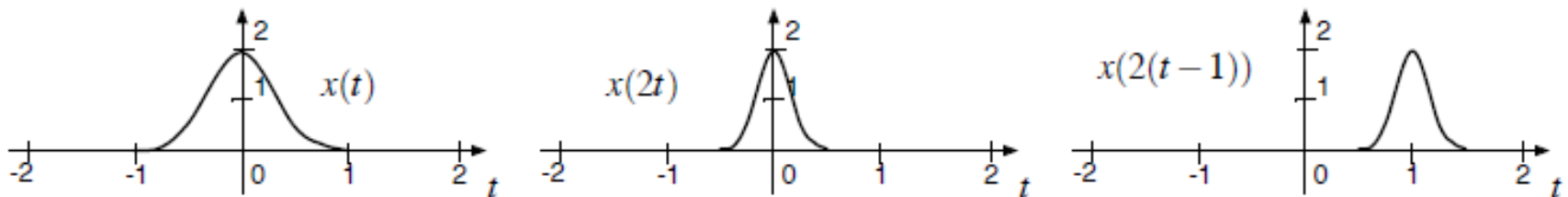
- Again, where is $n - n_1$ zero?

Combinations of Operations

- Time scaling, shifting, and reversal can all be combined.
- Operation can be performed in any order, but care is required.
- This *will* cause confusion.
- Example: $x(2(t - 1))$

Scale first, then shift

Compress by 2, shift by 1

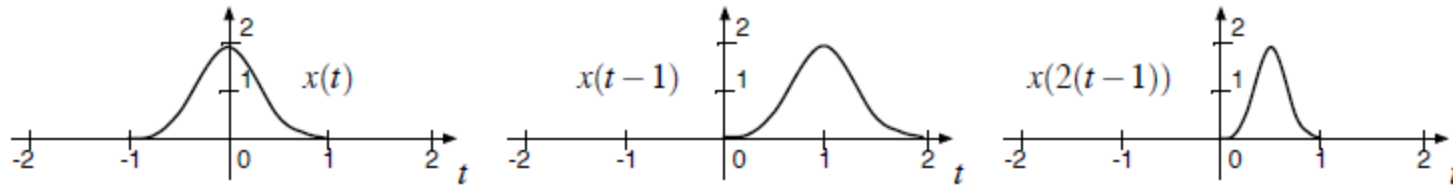


Combinations of Operations

Shift first, then scale

Shift by 1, compress by 2

Incorrect

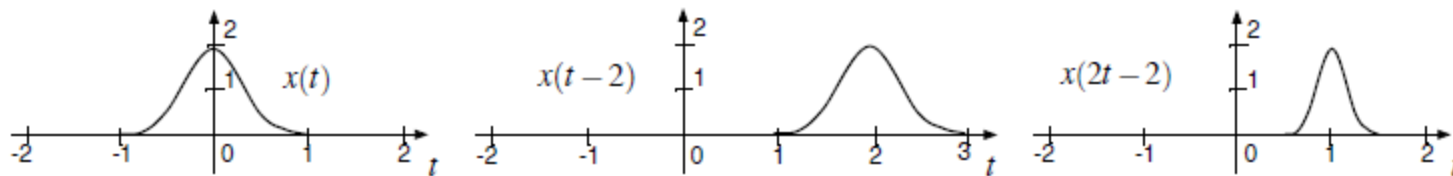


Shift first, then scale

Rewrite $x(2(t-1)) = x(2t-2)$

Shift by 2, scale by 2

Correct

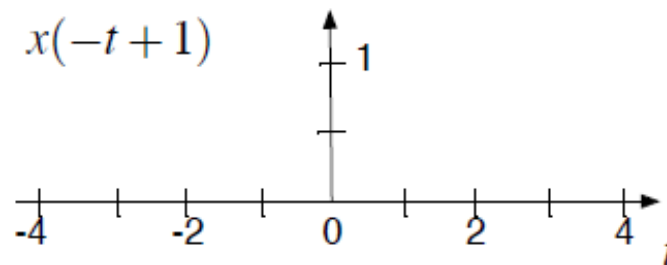
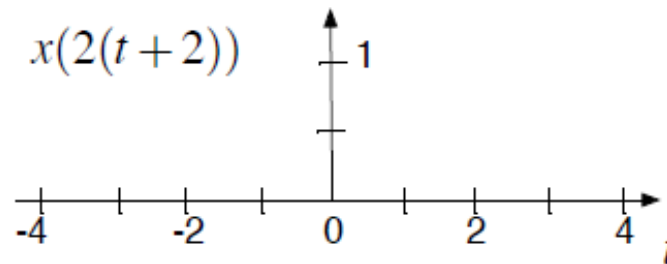
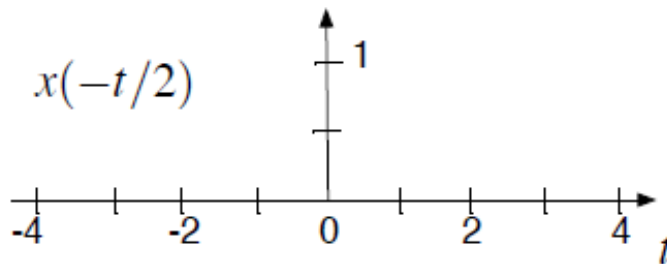
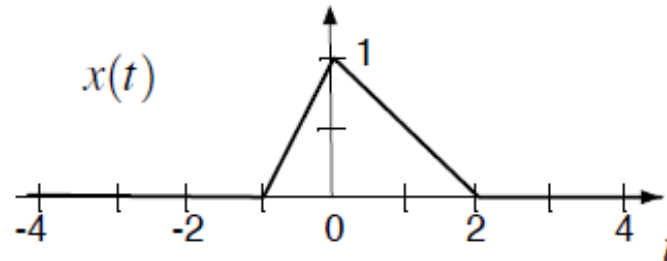


Where is $2(t-1)$ equal to zero?



Combinations of Operations

Try these yourselves



Review of Complex Signals

- Signals can also be complex

$$z(t) = x(t) + jy(t)$$

where $x(t)$ and $y(t)$ are each real valued signals, and $j = \sqrt{-1}$.

- Arises naturally in many problems
 - ▶ Convenient representation for sinusoids
 - ▶ Communications
 - ▶ Radar, sonar, ultrasound

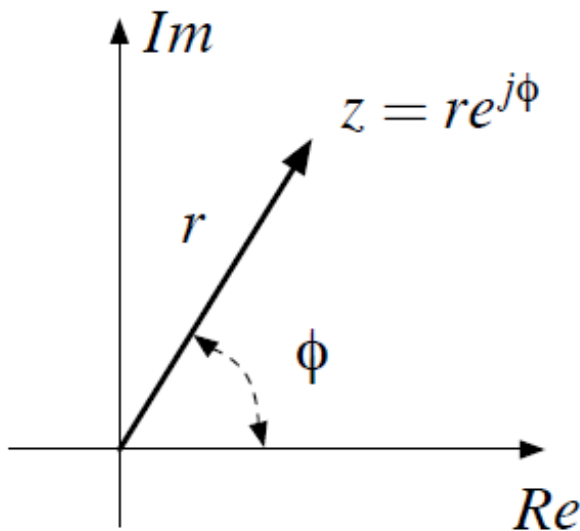
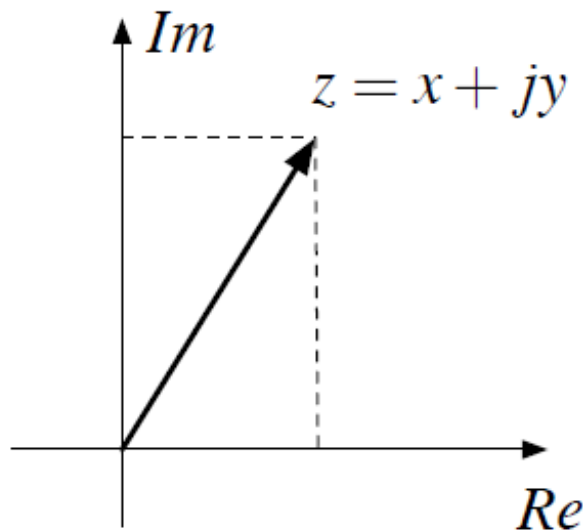
Complex number in Cartesian form: $z = x + jy$

- $x = \Re z$, the *real part* of z
- $y = \Im z$, the *imaginary part* of z
- x and y are also often called the *in-phase* and *quadrature* components of z .
- $j = \sqrt{-1}$ (engineering notation)
- $i = \sqrt{-1}$ (physics, chemistry, mathematics)

Review of Complex Signals

Complex number in polar form: $z = re^{j\phi}$

- r is the *modulus* or *magnitude* of z
- ϕ is the *angle* or *phase* of z
- $\exp(j\phi) = \cos \phi + j \sin \phi$



- complex exponential of $z = x + jy$:

$$e^z = e^{x+jy} = e^x e^{jy} = e^x (\cos y + j \sin y)$$

Know how to add, multiply, and divide complex numbers, and be able to go between representations easily.